REPORT ON MOVIE RATINGS ANALYSIS

1. Introduction

The movie industry produces thousands of films every year, and analyzing movie data helps identify patterns in audience preferences, movie performance, and industry trends. With the growth of data science, tools like Python provide powerful libraries that allow analysts to process and visualize large datasets efficiently.

In this project, Python libraries such as NumPy, Pandas, and Matplotlib are used to analyze a movie dataset. These libraries help perform statistical calculations, manipulate structured data, and create visualizations that reveal meaningful insights from the dataset.

The dataset contains information about movies such as title, genre, director, release year, budget, revenue, and ratings. By analyzing this data, we can understand how different factors like genre, director, and production budget influence movie ratings and overall performance.

This project demonstrates how data analysis techniques can be applied to real-world datasets to extract useful insights and visualize trends in the movie industry.

2. Objective

The main objective of this project is to analyze movie ratings using Python data analysis libraries and visualize trends in the dataset.

The specific objectives include:

- To load and explore a movie dataset using Pandas.
- To clean the dataset by handling missing or inconsistent values.
- To perform statistical analysis on movie ratings using NumPy, including calculating mean, median, and standard deviation.
- To analyze movie ratings based on different categories such as genre, director, and year using Pandas groupby operations.
- To create visualizations using Matplotlib to better understand the patterns and relationships within the dataset.
- To analyze relationships between budget, revenue, and movie ratings using scatter plots.
- To identify top-rated movies based on their ratings.

3. Dataset Description

- **Title:** The name of the movie.
- **Genre:** The category or type of the movie such as Action, Comedy, Drama, Thriller, etc.
- **Director:** The name of the director who directed the movie.
- **Year:** The year in which the movie was released.
- **Rating:** The average rating given to the movie by viewers, usually on a scale (for example 1–10).
- **Budget:** The total amount of money spent on producing the movie.
- **Revenue:** The total earnings generated by the movie from box office collections and other sources.

4. Data Cleaning and Preprocessing

In this step, the dataset was cleaned and prepared for analysis using Pandas.

First, the dataset was loaded using pd.read_csv() and inspected using head(), info(), and isnull().sum() to understand the structure and check for missing values. The dataset contains 18 rows and 7 columns: Title, Genre, Director, Year, Rating, Budget, and Revenue.

To ensure data quality, rows with missing Rating values were removed using dropna(). Missing values in Budget and Revenue were filled with 0 using fillna().

After cleaning, the dataset was ready for further statistical analysis and visualization.

5. Exploratory Data Analysis (EDA)

5.1 Descriptive Statistics of Movie Dataset

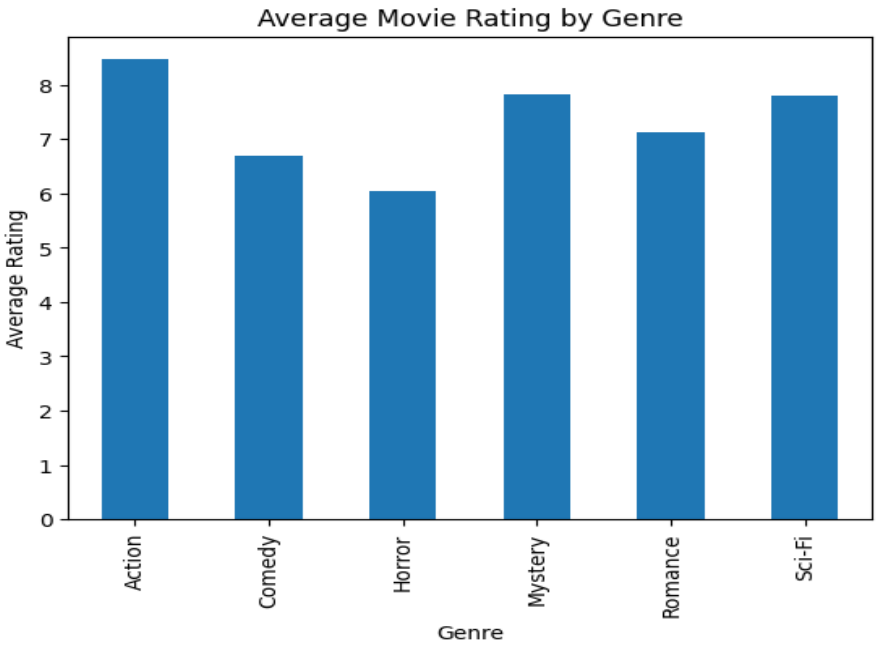
Findings:

- Average Rating: 7.33
- Median Rating: 7.45
- Rating Standard Deviation: 0.83
- Average Budget: 56.11
- Median Budget: 30.0
- Budget Standard Deviation: 58.12
- Average Revenue: 168.06
- Median Revenue: 110.0
- Revenue Standard Deviation: 146.14

Insight:

- Movie ratings are relatively stable around **7–8**, indicating generally positive audience feedback.
- Budget values vary significantly, suggesting that some movies are produced with much larger investments.
- Revenue also shows large variation, meaning some movies perform significantly better at the box office.

5.2 Average Rating by Genre



Findings:

- Action movies have the highest average rating (**8.47**).
- Sci-Fi (**7.80**) and Mystery (**7.83**) also have high ratings.
- Romance movies have moderate ratings (**7.13**).
- Comedy (**6.70**) and Horror (**6.03**) have the lowest average ratings.

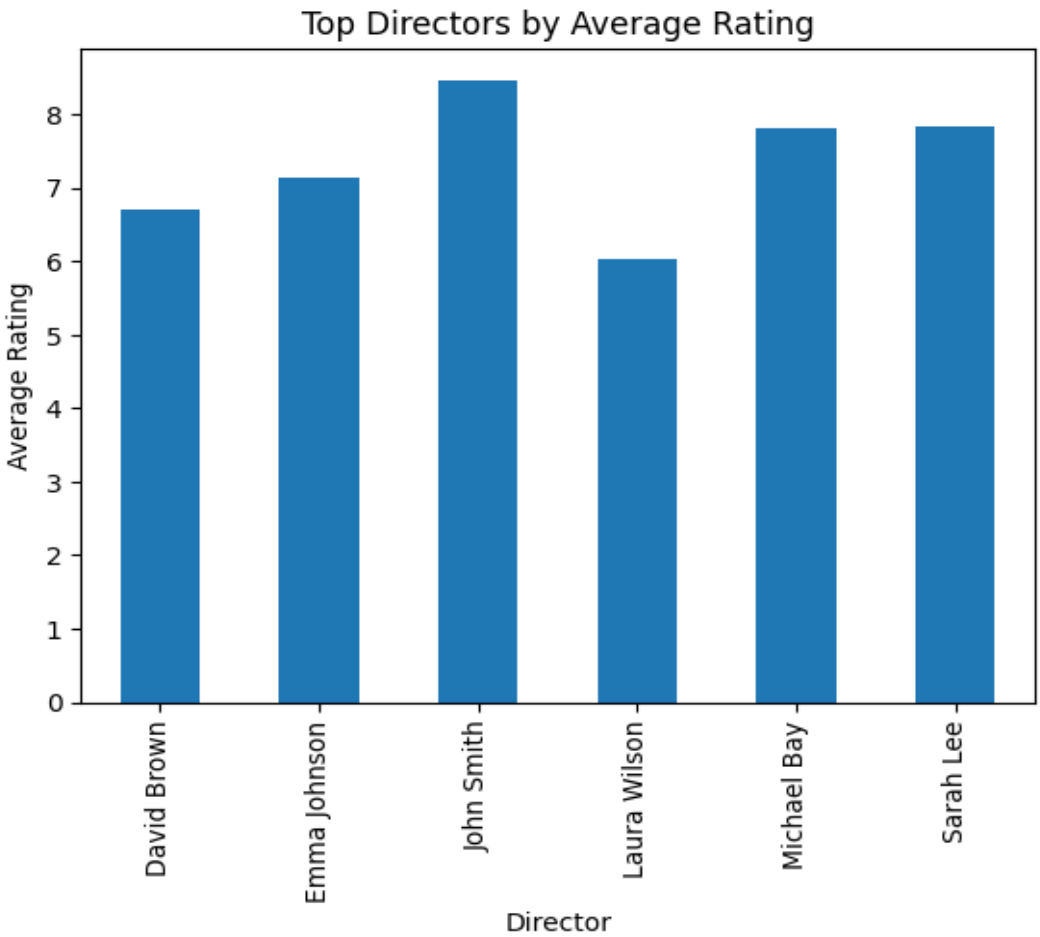
Insight:
The analysis shows that Action, Mystery, and Sci-Fi movies tend to receive higher audience ratings, while Comedy and Horror movies generally receive lower ratings in the dataset.

5.3 Average Rating by Director

Findings:

- John Smith has the highest average movie rating (8.47).
- Sarah Lee and Michael Bay also show strong ratings (7.83 and 7.80).
- Emma Johnson has moderate ratings (7.13).
- David Brown and Laura Wilson have comparatively lower ratings (6.70 and 6.03).

Insight:
Some directors consistently produce higher-rated movies, indicating that the director's style and experience may influence movie success and audience satisfaction.

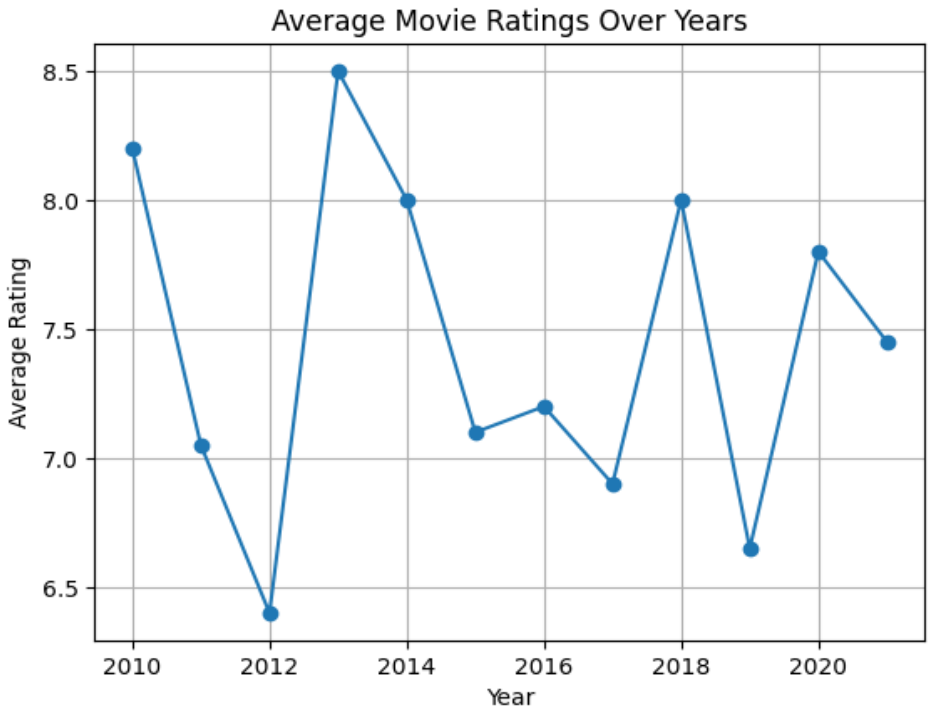


5.4 Average Movie Rating Trend Over Years

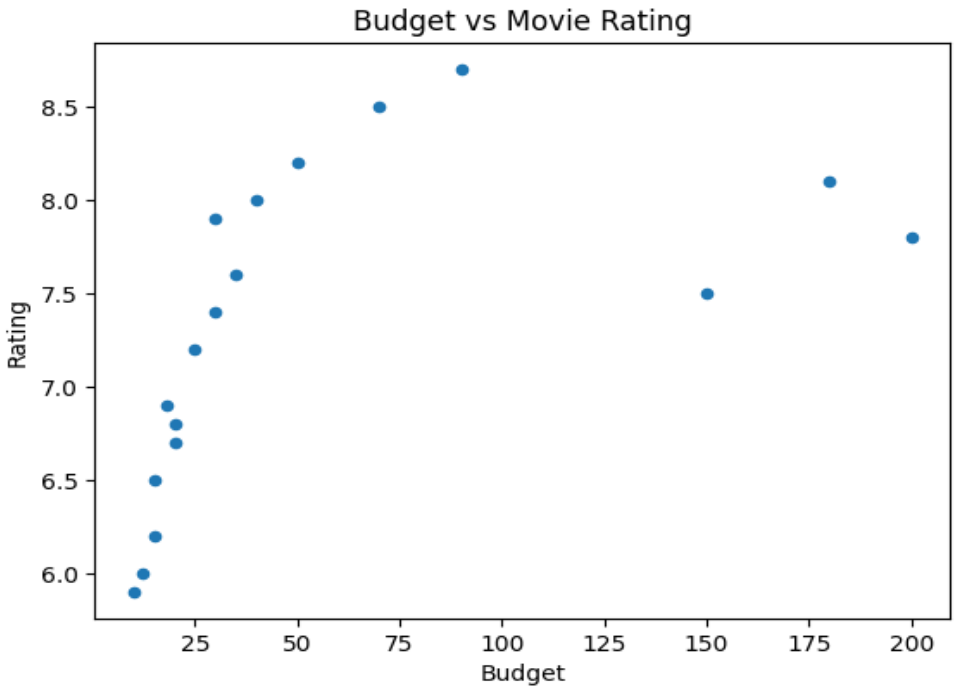
Findings:

- The highest average rating occurred in 2013 (8.5).
- High ratings were also observed in 2010, 2014, and 2018.
- Lower ratings occurred around 2012 and 2019.
- Ratings generally fluctuate between 6.5 and 8.5 over the years.

Insight:
Movie ratings vary across years, suggesting changes in movie quality, audience expectations, or genre trends over time.



5.5 Budget vs Movie Rating



Findings:

- Movies with higher budgets often have ratings between **7 and 8.5**.
- Some low-budget movies also achieve good ratings.
- The relationship between budget and rating is not perfectly linear.

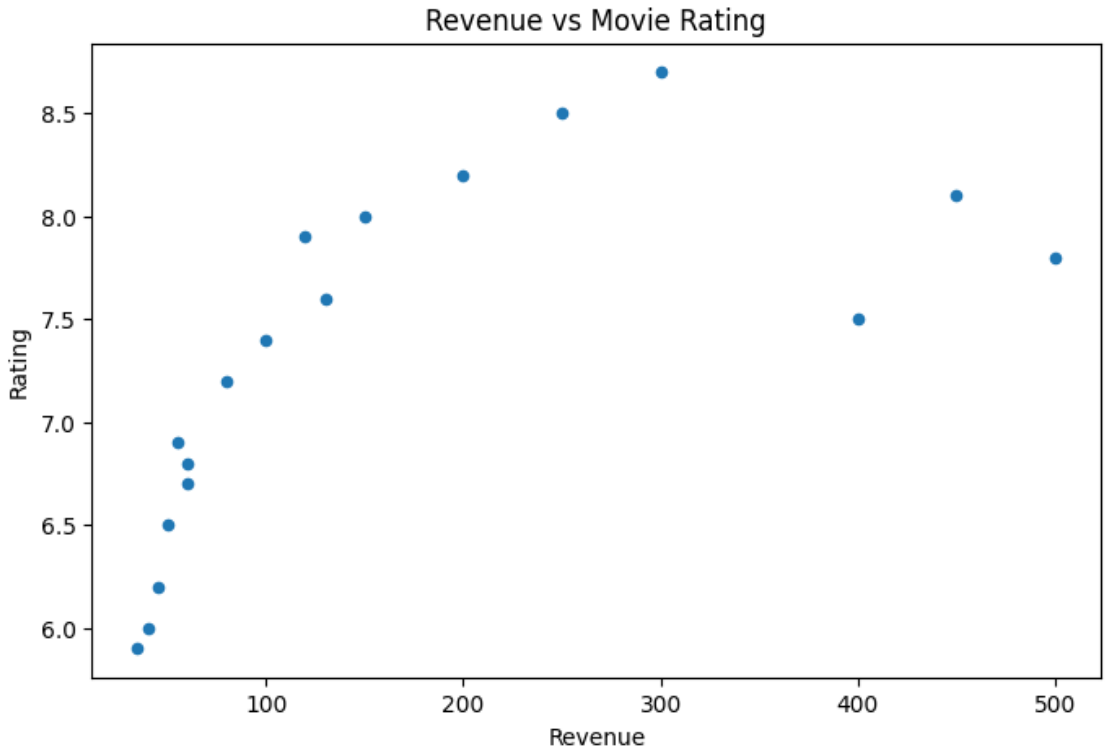
Insight:
Higher budgets may help improve production quality, but a bigger budget does not always guarantee higher ratings.

5.6 Revenue vs Movie Rating

Findings:

- Movies with higher ratings often generate higher revenue.
- Some movies with moderate ratings still achieve strong revenue.
- Revenue values range widely from around 40 to over 500.

Insight:
There is a moderate positive relationship between ratings and revenue, suggesting that well-rated movies often perform better commercially.

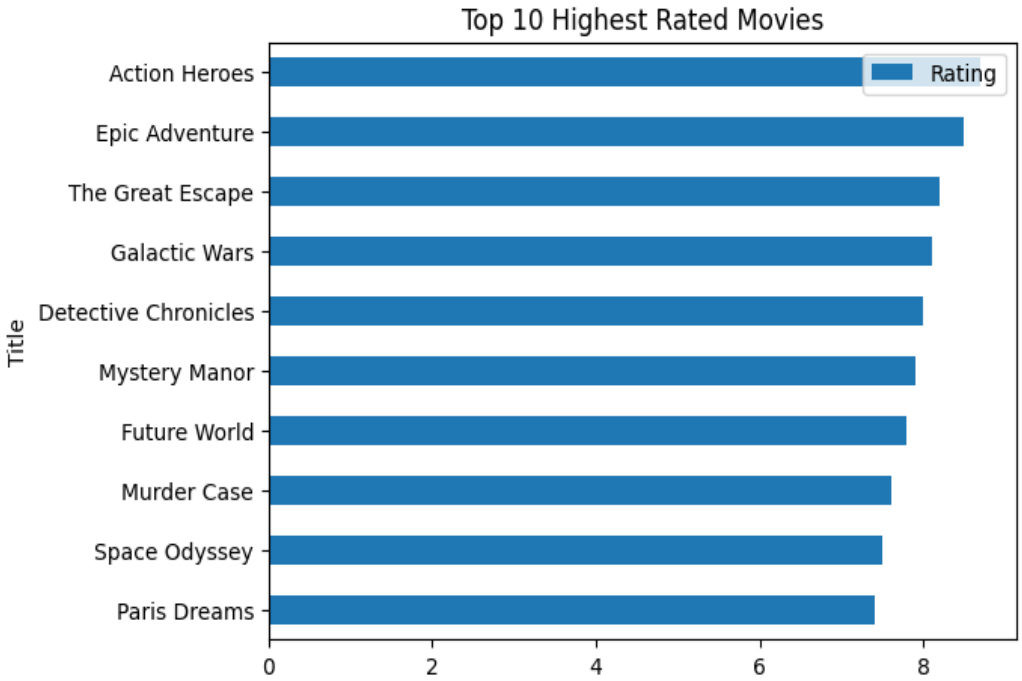


5.7 Top Rated Movies

Findings:

- The dataset includes several movies with ratings above **8.0**.
- Action heroes is the **highest-rated movies**.

Insight:
Highly rated movies tend to stand out clearly in the dataset and may represent successful films in terms of audience satisfaction and overall quality.



6. Key Insights

• The average movie rating in the dataset is **7.33**, indicating that most movies receive generally positive audience feedback.

• **Action, Mystery, and Sci-Fi** genres have the highest average ratings, suggesting that audiences prefer movies from these genres.

• Among the directors, **John Smith** has the highest average movie rating, indicating strong audience appreciation for movies directed by him.

• Movie ratings fluctuate across years, with the highest average rating observed in **2013**.

• Movies with higher budgets often achieve good ratings, but **a large budget does not always guarantee higher ratings**.

• There is a **moderate positive relationship between revenue and ratings**, meaning highly rated movies tend to perform better at the box office.

• The analysis also highlights several **top-rated movies with ratings above 8**, showing strong audience satisfaction.

7. Conclusion

This project analyzed a movie dataset using NumPy, Pandas, and Matplotlib to understand trends in movie ratings and relationships between different variables. The dataset was first cleaned and inspected to ensure data quality before performing statistical analysis and visualizations.

The analysis revealed meaningful insights about genre popularity, director performance, rating trends over time, and the relationship between budget, revenue, and movie ratings. Visualizations such as bar charts, line graphs, and scatter plots helped present the findings clearly.

Overall, the project demonstrates how Python data analysis tools can be used to explore datasets, identify patterns, and generate useful insights. These techniques can be applied to larger movie datasets to support decision-making in the film industry and understand audience preferences more effectively.